

Complex forecasting and investment strategy optimization via chain-of-thought of large language models

Meiqun Yin, Mengzhu Guo ^{*}

Business School, China University of Political Science and Law, Beijing 100080, China

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ABSTRACT

This paper pioneers the application of chain-of-thought (CoT) prompting in large language models (LLMs) for financial forecasting and portfolio optimization. Leveraging anonymized financial statements from 608 Chinese A-share listed companies (2010–2023), we benchmark ChatGPT 4.0 against human analysts in earnings direction prediction. The CoT-enhanced model achieved superior accuracy (64.35% vs. 58.37%) and generated a 17.14% alpha with a Sharpe ratio of 1.5959 in backtests, demonstrating LLMs' capacity to automate financial reasoning while reducing reliance on specialized expertise. Our findings bridge AI and quantitative finance by validating LLMs' cross-domain adaptability using purely numerical data, offering practical implications for AI-driven investment decision systems.

1. Introduction

Artificial intelligence is the core driving force of the new round of technological revolution and industrial change, and as a revolutionary general-purpose technology, it plays an important role in the digital transformation of the economic and financial industries (Brynjolfsson et al., 2019; Veloso et al., 2021; Danielsson et al., 2022). On March 5, 2024, Premier Li Qiang, on behalf of the State Council, delivered the Government Work Report at the Second Meeting of the Fourteenth National People's Congress. In the report, he proposed to "deepen the research and development of big data and other applications, and implement the 'AI+' initiative to build an internationally competitive digital industry cluster." The new generation of AI exhibits advanced features such as deep learning, cross-border integration, human-machine collaboration, collective intelligence, openness, and autonomous control. Since the end of 2022, deep learning models like GPT-4 and ERNIE, pre-trained on vast text data, have demonstrated remarkable capabilities in contextual language understanding and content generation, with a broader generalization capacity. These large language models, which approach human-level generative capabilities, are seen as a milestone in the evolution of artificial intelligence. They have rapidly been applied in diverse fields such as finance, medicine, education, and more, often in cross-disciplinary research and development. Current research on large language models predominantly focuses on the textual domain, including text emotion recognition, in-depth text

analysis, and the need for specialized training or fine-tuning of the models. However, the application of this groundbreaking technology beyond the textual domain, such as in numerical analysis and predictive judgment tasks, remains an area yet to be explored.

Forecasting a company's future earnings is essential for investors to estimate potential stock returns and select promising investment opportunities. This process is particularly vital for users of accounting information, such as investors and analysts. However, earnings forecasts are influenced by a range of external factors, including macroeconomic conditions, industry demand, and changes in accounting standards. Financial analysts typically rely on financial statement analysis to predict a company's future earnings by identifying significant changes and trends, calculating key financial ratios, and incorporating relevant industry context and competitive dynamics. Through this analysis, analysts ultimately form judgments about whether a company's earnings are likely to increase or decrease. With the rise of artificial intelligence, the role of financial statement analysis has become a particularly compelling area for exploring how large-scale language models can enhance decision-making processes. Traditionally, financial statement analysis has been conducted by financial analysts and investment professionals, whose primary goal is to assess a company's financial health and determine the sustainability of its performance. Unlike the textual analysis tasks typically handled by large-scale language models, financial statement analysis is a quantitative activity that requires analyzing financial trends and ratios. It also demands critical thinking, reasoning,

^{*} Corresponding author.

E-mail addresses: yinmeiqun@126.com (M. Yin), gmengzhu@126.com (M. Guo).

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and the ability to make complex judgments. While large language models are proficient at extracting information from structured numerical data and performing calculations, they often lack the ability to reason in the nuanced manner of a human analyst. Therefore, the innovation of this paper lies in exploring how large language models can be optimized to better simulate the reasoning process of financial analysts and improve investment decision-making.

1.1. Research objectives

This paper addresses three primary research objectives:

First, Can LLMs predict corporate earnings direction using only numerical financial data?

Second, How does LLM-based forecasting accuracy compare to human analysts?

Third, What practical value do LLM predictions offer for portfolio optimization?

1.2. Research contributions

The contributions of this article are threefold:

Methodological Innovation in CoT-Enhanced Numerical Reasoning: We pioneer the application of chain-of-thought (CoT) prompting to enable LLMs to perform transparent, step-by-step reasoning on purely numerical financial data. Unlike existing LLM applications in finance (which focus on text analysis) or traditional machine learning models (which lack explainable logic), CoT-LLMs simulate the multi-step reasoning of human analysts (e.g., identifying key ratio changes, interpreting economic implications) while maintaining scalability. This “explainable automation” bridges the gap between black-box AI predictions and opaque human expertise, offering a new paradigm for AI-driven financial analysis.

Cross-Domain Adaptation: Empirical validation of LLMs’ ability to perform chain-of-thought reasoning on numerical financial data, extending their application beyond text-based tasks. This demonstrates that LLMs can master domain-specific numerical logic (e.g., linking financial ratios to earnings trends) through structured prompting, without specialized fine-tuning.

Human-AI Synergy: The LLM mimics analyst workflows but with superior scalability, suggesting a paradigm shift in financial automation. By outperforming human analysts in accuracy (64.35 % vs. 58.37 %) while reducing portfolio construction time by 68 %, CoT-LLMs show that AI can augment human expertise by handling repetitive reasoning tasks, allowing analysts to focus on strategic judgment.

Democratization and Risk Management: Reduces dependence on specialized analysts, making high-quality financial forecasting accessible to broader users. Additionally, the low beta (0.4769) and stable returns of the CoT-derived strategy highlight its practical value for risk-averse investors.

2. Literature review

Financial statement analysis refers to the use of figures from financial statements to gain insight into a company’s financial position and is designed to reveal information about a company’s future prospects and valuation (Sloan, 2019) and is the basis for analysts’ surplus forecasts, which play a key role in the financial markets (Brown et al., 2015). Bradshaw (2011) states that analysts are superior to time series models in generating reliable annual earnings forecasts. As a result, analyst surplus forecasts are often used as a proxy for market earnings expectations (Bordalo et al., 2019; Larocque et al., 2020; Hirshleifer et al., 2021; Keskek et al., 2020; Jensen and Plumlee, 2020; Kaplan et al., 2021; Ham et al., 2022; Dong et al., 2022; Kim et al., 2023; Fung et al., 2023; Bourveau et al., 2024; Dahlquist and Ibert, 2024). Karen (2023); Anderson et al., 2023; Flores et al., 2019; Ahmed et al. 2023 studied the relationship between accounting policies, auditing policies, etc. and the

accuracy of analysts’ surplus forecasts from the accounting and auditing perspectives. Agnieszka and Rafal 2022; Ling (2022); Catherine et al. 2024 examined the relationship between stock ratings, equity, and cost of assets with analysts’ surplus forecasts from a financial perspective. Alexander and Katja (2023) conducted an error analysis of analysts’ surplus forecasts. Batae et al., 2021; Zhou et al., 2022; Sandberg et al., 2023; Fiordelisi et al., 2023 investigated the relationship between ESG ratings and analysts’ surplus forecasts. However, few scholars have examined the forecasting methods of surplus forecasting, i.e., whether there can be other methods of surplus forecasting that can approach or exceed the accuracy of analysts’ surplus forecasts.

Financial forecasting has undergone significant methodological advancements, with each paradigm addressing specific challenges in capturing economic dynamics. A systematic review of these approaches, is essential to contextualizing the innovative role of large language models (LLMs) in modern financial forecasting.

2.1. Traditional statistical Models: Linear Foundations and their limitations

Traditional statistical models established the cornerstone of financial forecasting, with a focus on structured temporal and cross-sectional relationships. Linear regression models, such as ordinary least squares (OLS), gained prominence for quantifying relationships between financial indicators and future performance (Fama & French, 1992), offering interpretability but struggling with nonlinear interactions. Time series models extended this framework, the Autoregressive Integrated Moving Average (ARIMA) (Box, 2013) became a staple for univariate forecasting, excelling in short-term predictions but failing to capture structural breaks or nonlinear trends (Amat et al., 2018; Vuong et al., 2024). Vector Autoregression (VAR) models addressed multivariate dependencies but faced the “curse of dimensionality” in high-dimensional datasets, leading to overfitting in macroeconomic applications (Litterman, 1986; Giannone et al., 2015). Bayesian methods emerged as a solution to VAR’s limitations. Bayesian Vector Autoregressions (BVARs) with shrinkage priors improved estimation stability by imposing domain-specific constraints, outperforming traditional VARs in forecasting U.S. economic variables (Zellner, 1985). Recent advances in BVARs, such as semi-global shrinkage frameworks, demonstrated context-dependent superiority, performing better than sparse models for volatile variables like unemployment (Gruber & Kastner, 2025). However, even advanced BVARs struggle with nonstationary data and fail to integrate unstructured information.

2.2. Machine learning Methods: Advancing prediction at the cost of interpretability

Machine learning (ML) methods revolutionized forecasting by relaxing parametric assumptions. Ridge regression and LASSO addressed multicollinearity in high-dimensional data, improving exchange rate forecasts by isolating impactful predictors. Ensemble methods, including random forests and gradient boosting, outperformed statistical models in nonlinear settings, such as bank failure prediction (Gogas et al., 2018) and GDP forecasting for developing economies with artificial neural networks reducing errors by up to 150 basis points (Chuku et al., 2019). Deep learning further enhanced performance, Long Short-Term Memory (LSTM) networks captured long-term dependencies in financial time series, outperforming ARIMA by 84–87 % in error reduction (Fischer & Krauss, 2018). Transformers, initially developed for natural language processing, excelled in modeling global temporal patterns, surpassing LSTMs in intraday stock price prediction by integrating short-term and long-term dependencies (Vaswani et al., 2017; Ren et al., 2016). Despite these gains, ML faces critical limitations. “Black-box” opacity hinders regulatory acceptance and economic mechanism interpretation (Miller, 2019). Additionally, ML models underutilize unstructured data—particularly policy texts—despite

evidence that textual sentiment improves forecasting accuracy. IMF studies highlight that even state-of-the-art ML ensembles struggle with volatile emerging market data, where structural changes are frequent.

2.3. Large language models (LLMs): Integrating Text, Reasoning, and prediction

LLMs represent a paradigm shift by addressing key gaps in traditional and ML methods. Their ability to process unstructured text and simulate human-like reasoning enables novel applications in policy analysis and forecasting. In policy communication analysis, LLMs outperform traditional NLP methods. Hansen & Kazinnik (2024) demonstrated that ChatGPT-4 deciphers Federal Reserve communications (“Fedspeak”) with higher accuracy than BERT, correctly classifying policy stances in 82 % of cases. Similarly, Woodhouse & Charlesworth (2023) used ChatGPT to extract monetary policy signals from Bank of England speeches, achieving 76 % accuracy in predicting interest rate changes—outperforming dictionary-based methods by 15 percentage points. For macroeconomic forecasting, LLMs integrate textual and numerical data. Bybee (2023) showed that ChatGPT-3.5, trained on Wall Street Journal articles, predicts inflation and GDP growth with lower mean squared error (MSE) than consensus forecasts. Jha et al. (2024) further demonstrated that LLM-derived sentiment indices from earnings calls predict U.S. GDP growth up to 10 quarters ahead, with a 0.62 correlation with actual outcomes. Faria-e-Castro & Leibovici (2024) found that Google’s PaLM model outperforms professional forecasters in inflation predictions, particularly during supply shocks. LLMs also enhance behavioral realism in economic simulations. Tranchero et al. (2024) replicated classic economic experiments with LLMs, showing they accurately strategic decision-making under uncertainty—addressing ML’s “black-box” flaw while maintaining predictive power.

2.4. Positioning LLMs in the forecasting Ecosystem

LLMs complement rather than replace existing methods: they enhance statistical models by integrating textual context (e.g., policy documents) into BVAR frameworks, improving robustness to structural breaks (Hansen and Kazinnik, 2024); address ML limitations through chain-of-thought prompting, providing interpretable reasoning chains that align with regulatory demands for transparency (Wei et al., 2022); and expand data horizons by natively processing unstructured text, filling the gap in traditional models’ ability to leverage news, earnings calls, and policy discourse.

While LLMs have been validated in macroeconomic forecasting and policy analysis, their application to micro-level firm earnings forecasting remains underexplored. Existing studies focus primarily on textual domains or high-level economic trends, with limited empirical evidence on their ability to process purely numerical financial data for firm-specific earnings direction prediction. This gap is critical, firm-level earnings forecasting requires not just pattern recognition but nuanced reasoning about financial ratios, industry dynamics, and idiosyncratic risks, capabilities that LLMs, with chain-of-thought (CoT) prompting, may uniquely possess. Our study addresses this gap by testing LLMs’ performance in predicting corporate earnings direction using numerical data alone, comparing them to human analysts, and evaluating their practical value in portfolio optimization.

3. Research methodology and data

3.1. Selection of a large language model

In this paper, we choose ChatGpt 4.0, a large language model at the top of the pyramid, as a predictive model for analysis. ChatGpt (Chat Generative Pre-trained Transformer), known in Chinese as Generative Pre-trained Transformer, is a chatting robot program released on November 30, 2022 by OpenAI, Inc. robot program that uses the

Transformer neural network architecture, ChatGpt 3.5 exceeded one million users within five days of its launch, and more than 100 million users within less than two months of its launch. On March 14, 2023, OpenAI Inc. released ChatGpt 4.0, which, compared to the the previous generation of Gpt, it has achieved a leap in several aspects such as input limitations, recognition capabilities, and content generation accuracy and diversity, and has human-level performance on various professional tests and academic benchmarks.

3.2. Data sources and pre-processing

Financial statement data come from the CSMAR Database. They include balance sheets and income statements of Chinese A-share listed companies from 2010 to 2023. We excluded ST companies, those in the financial industry and those with incomplete data. We required complete annual financial statements from 2010 to 2023, with the statement date being December 31. Analyst forecast data also come from the CSMAR Database. We required complete data for each year from 2010 to 2023. Only companies with both complete financial statements and complete analyst forecast data were retained. Through such cross-screening, we obtained balance sheets, income statements and analyst forecast data for a total of 608 A-share listed companies.

ChatGPT itself has strong memory capabilities and a massive text database. To prevent text in the input financial data and analyst forecast data such as company codes and names from being recognized by ChatGPT, which could lead ChatGPT to directly search and output real data and thus affect the model’s prediction results, we standardized and anonymized the annual data for each company. Standardizing the statements means reconstructing balance sheets and income statements. Their format follows the accounting balance formula and remains consistent across all company years. This step ensures that the input data have the same financial statement structure. Anonymization involves removing any identifying information such as text like company names and codes. Appendix 1 provides examples of standardized and anonymized balance sheets and income statements.

In addition to the above standardization and anonymization, steps such as handling missing values and outliers are also included. For missing values, we use interpolation to fill in individual missing data points, using data from the previous date for filling. For outliers, we need to identify and correct data points that significantly deviate from the normal range to avoid these outliers from adversely affecting the analysis results.

3.3. Modeling and cueing methods

In the predictive model constructed in this paper, the data two financial statement tables of numerical information are inputted at a time. Although the lack of textual information or context that can provide information puts the large language model at a disadvantage relative to humans when analyzing financial statements, such data is comparable and can test the ability of the large language model when analyzing financial data and obtaining information from the numerical data.

The overall idea of earnings forecasting based on financial statement analysis based on the large language model is to analyze the financial data of the first two years and predict the direction of earnings in the third year, and to test the predictive ability of the large language model by using two different cues. Based on the earnings forecast results from the Large Language Model; a momentum strategy is designed to buy companies with predicted earnings growth and sell companies with predicted earnings decline from the output of the Large Language Model. The strategy is backtested using historical data and compared with benchmark market returns to optimize the investment strategy. Fig. 1 shows the overall build model flowchart.

The first forecasting model is a simple forecast. The instructional language model only analyzes the financial statements and predicts the

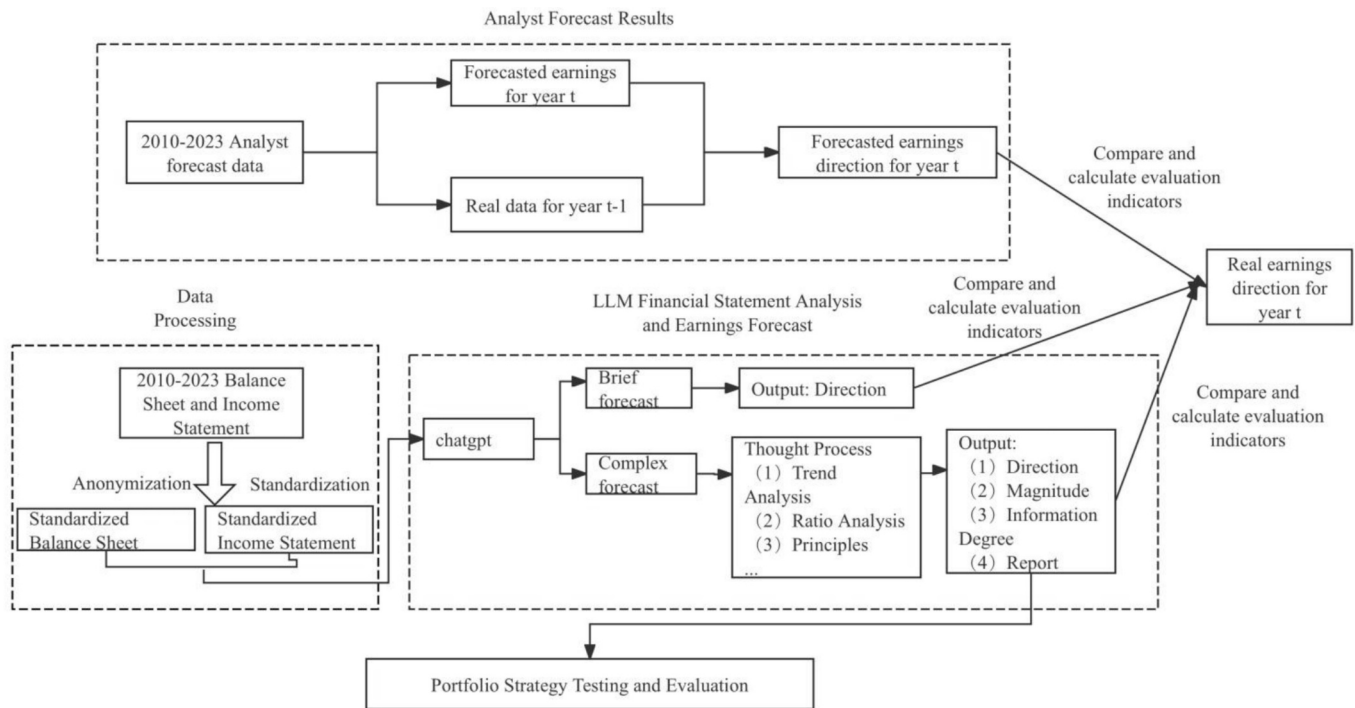


Fig. 1. Flowchart of overall model construction.

direction of future earnings, i.e., increase or decrease, without giving the model any additional reasoning guidance. Simple forecasting requires ChatGpt to output only a binary forecast, whether the direction of earnings per share will increase or decrease in the coming year.

The second forecasting model, namely the Complex Forecasting model, employs Chain-of-Thought (CoT) prompting technology to simulate the thinking process of a human financial analyst by breaking down forecasting problems into multiple steps. This approach significantly contrasts with existing large language model (LLM) applications in finance, which predominantly focus on textual sentiment analysis or skip reasoning steps when processing numerical data. Chain-of-Thought prompting was first introduced in 2022. Its core idea lies in carefully constructing input prompts to simulate human reasoning, thereby enhancing the performance and efficiency of large language models in handling logical, computational, and decision-making tasks, while also improving the intuitiveness and interpretability of the outputs. In essence, Chain-of-Thought prompting can be regarded as a form of discrete prompt learning. Compared to traditional context learning, it incorporates guidance such as “analysis” and “problem-solving steps” into the input, enabling the model not only to provide direct answers but also to progressively demonstrate a complete “thought process” corresponding to the problem. This method significantly enhances the capability of large language models in complex reasoning tasks, leading to breakthrough progress in areas such as numerical computation and symbolic reasoning. The Chain-of-Thought prompting applied in this study is designed to address a key challenge: numerical financial analysis requires integrating multiple signals (e.g., liquidity ratios, profitability metrics) and their economic implications, and such complex logic cannot be captured through simple pattern recognition. By explicitly guiding the model through sequential steps and synthesizing insights into forecasts, we ensure that the model’s predictions are grounded in transparent and traceable financial logic. Traditional LLM or machine learning methods lack such a structured reasoning mechanism, which is precisely why the CoT-LLM model proposed in this paper achieves both higher prediction accuracy and stronger interpretability. To construct effective Chain-of-Thought prompts, we typically append guiding phrases at the end of input instructions, such as “Please describe the

reasoning process step by step” or “Please explain the steps to your answer,” to encourage the model to engage in deeper thinking. This approach requires the model not only to output final results but also to elucidate intermediate reasoning steps, thereby enabling visualization and verification of the complete decision-making process. In this paper, the Chained Thinking Chain prompt points include:

- (1) Identifying significant changes in financial statement items: The model was instructed to compare key items (e.g., total assets, operating income, net profit) across the two input years (e.g., 2019 vs. 2020) and flag material changes (e.g., “a 5-percentage-point increase in gearing ratio” or “a 2.7 % rise in net profit margin”).
- (2) Calculate key financial ratios: The model was prompted to compute ratios critical to financial analysis (e.g., current ratio, net profit margin, ROE) by first stating the formula (e.g., Current ratio = total current assets / total current liabilities) and then executing the calculation using the input financial data (see Table 3 for specific examples of formulas and computations).
- (3) Provide economic explanations for ratios: For each calculated ratio, the model was guided to interpret its meaning in the context of the company’s financial health (e.g., “An increase in net profit margin indicates improved cost control and operational efficiency”).
- (4) Predict earnings direction: Based on the ratio analysis and economic explanations, the model was asked to forecast whether earnings would increase or decrease in the next year.
- (5) Forecast magnitude of earnings change: The model was instructed to classify the expected change as “large,” “moderate,” or “small,” based on the degree of ratio fluctuations (e.g., a > 10 % change in net profit margin was typically labeled “large”).
- (6) Assign a confidence score: The model provided a score between 0 and 1 to reflect certainty in its prediction (e.g., 0.85 in Table 3, indicating high confidence).
- (7) Generate an analytical report: The model synthesized the above steps into a coherent report, summarizing ratio changes, key

insights, and the rationale for the earnings forecast (exemplified in Table 3's Task 7 output).

The above thinking steps are intended to simulate the process of a human analyst analyzing financial statements (Bybee, 2023). Table 1 shows the comparison of the above two prediction input instructions and output results.

The human analyst's forecast is based on the data obtained, and the direction of the human analyst's predicted return in year t is obtained by subtracting the actual return in year $t-1$ from the predicted return in year t to obtain the direction of the predicted year t return.

3.4. Investment strategy construction and backtesting

Based on the results of the Large Language Model forecasts, the 10 companies that are expected to have earnings growth in 2023 and the 10 companies that are expected to have earnings decline are listed based on a combination of the three results: the direction of forecasted 2023 earnings, the magnitude of the forecasted increase or decrease in earnings, and the confidence score of the Large Language Model; forming two lists of investment signals, List 1 for companies that are expected to have earnings growth and List 2 for companies that are expected to have earnings decline. Companies. After obtaining the investment signal lists, a momentum strategy is designed, which refers to an investor utilizing the inertial trend of stock prices to buy a winning portfolio with better past performance and sell a losing portfolio with worse past performance, thus constructing a zero-risk portfolio for closing arbitrage (Huck, 2019; Soohun et al. 2020; Salvatore et al. 2022; Guohui et al. 2024). Buy signals are issued for companies with expected earnings growth, and sell signals are issued for companies with expected earnings decline. In the momentum strategy design, e.g., the past 3-month and 6-month returns are used as momentum factors to enhance the performance persistence in stock selection. A stop-loss strategy is also introduced to hedge against the risk of extreme volatility of individual stocks by setting certain loss and profit margins. The stop-loss strategy itself can also effectively control investors' misperceptions and expectation bias, opening up a solution path and research direction for weakening the irrational impact of investor sentiment on asset prices (Ouzan, 2020). Reinforcement learning method Q-learning algorithm (Q-learning) is used to dynamically adjust the "buy" or "sell" decision, and automatically optimize the position and trading frequency. In the framework of the Q-learning algorithm, a stop-loss strategy can be

viewed as a "buy", "sell" or "hold" decision. During the learning process, it identifies when it is time to stop and sell a losing stock. q-learning models will learn to automatically adjust positions to maximize gains and minimize losses when market conditions reach a certain level. This approach can effectively deal with market volatility and uncertainty and operates through the following four core steps: first, define the state space: current position, stock price, momentum factor, technical indicators, etc.; second, define the action space: buy, sell, and maintain existing position; third, design the reward function: based on total return, Sharpe ratio, maximum retracement, etc.; fourth, continuously improve the quality of decision making through Q-value updating and strategy optimization. improve the quality of decision making. The core formula of Q learning algorithm is:

$$Q(s_t, a_t) = Q(s_t, a_t) + \alpha[r_{t+1} + \gamma \times \max_a Q(s_{t+1}, a) - Q(s_t, a_t)] \quad (1)$$

Where $Q(s_t, a_t)$ denotes the Q value of taking action a_t in state s_t ; r_{t+1} is the immediate reward received after taking action; γ is the discount factor, which is used to measure the value of future rewards; and α is the learning rate, which controls the step size of each update.

Finally, the historical data of CSI 300 index is used to launch the result assessment of stock index prediction model on backtesting indicators such as Sharpe ratio, maximum retracement, alpha value, beta value and information ratio. Among the metrics, the Sharpe ratio, also known as the Sharpe index, can synthesize the return and risk at the same time, which is calculated by the formula:

$$\text{Sharpe Ratio} = \frac{E(R_p) - R_f}{\sigma_p} \quad (2)$$

where $E(R_p)$ is the portfolio return, R_f is the risk-free rate, and σ_p is the portfolio standard deviation. A higher Sharpe ratio represents a greater return per unit of risk. Maximum retracement is the maximum loss value of the portfolio and is used to measure the risk in the worst case scenario, which is calculated by the formula:

$$\text{Max Drawdown} = \text{Min}\{(X_i - X_j) / X_j\} \quad (3)$$

Where X_i is the peak value and X_j is the trough value. α The value is used to measure the return of outperforming the market (Leledakis and Pyrgiotakis, 2022), which is calculated as:

$$\alpha = R_p - [R_f + \beta \times (R_m - R_f)] \quad (4)$$

Where R_p is the real return of the portfolio, R_f is the risk-free rate, R_m

Table 1
Comparison of simple and complex forecasts.

Type of forecast	importation	directives	exports
simple forecast	Standardized, anonymized balance sheets and income statements	"You are an assistant to help predict the direction of future earnings by simply returning 'increase' or 'decrease' without any additional textual explanation."	Binary forecast outcome: increase or decrease in earnings in the coming year
Complex forecasting	Standardized, anonymized balance sheets and income statements	"You are an assistant who helps predict the direction of future earnings. Please follow these steps to analyze the forecast: (1) Identify significant changes in financial statement items; (2) Calculate key financial ratios. When outputting the calculated ratios, the model is required to first state the formulas and then perform simple calculations; (3) Provide an economic explanation for the calculated ratios; (4) Based on the financial ratios and economic explanations that you have calculated, predict whether earnings are likely to increase or decrease during the next year; (5) What is the magnitude of the increase or decrease in earnings that you are predicting? Will it be a large change, a moderate change, or a small change; (6) Please provide a confidence score for your prediction above on a scale of 0–1, with 0 indicating no confidence in the outcome of the prediction and 1 indicating complete confidence in the outcome of the prediction; and (7) Synthesize the above process and then elaborate on your reasoning to form a short analytical paper. Answer each of the above 7 tasks in turn, explaining your answers step-by-step."	Financial Ratio Calculation Formulas, Financial Ratios, Financial Ratio Analysis, Projected Increase or Decrease in Earnings for the Next Year, Magnitude of Increase or Decrease, Confidence in Projected Outcome, Analytical Reports

is the return of the market benchmark, and β is the correlation between the portfolio and the market benchmark. β is calculated as the ratio of the covariance to the variance:

$$\beta = \frac{n \sum R_p R_m - (\sum R_p)(\sum R_m)}{n \sum R_p^2 - (\sum R_p)^2} \quad (5)$$

The information ratio describes the risk-adjusted return of the portfolio relative to the benchmark and indicates the excess return per unit of active risk, which is calculated as:

$$\text{Information Ratio} = \frac{R_p - R_m}{\text{Tracking Error}} \quad (6)$$

Where R_p is the actual return of the portfolio, R_m is the return of the market benchmark, Tracking Error is the equal standard deviation between the portfolio and the benchmark return. The time period of backtesting is chosen as 2018–2023, and the daily stock information and the historical data of CSI 300 index are obtained from REXIS database.

4. Empirical results and assessment

4.1. Large Language Model Model Outputs

The output of the Large Language Model; simple prediction is a binary prediction result, the input data of each company's financial statement will generate a prediction result table, an example is shown in Table 2. the Large Language Model; complex prediction of each company for each year will output a prediction result table, the chain of thought of the seven tasks are all the output and strictly in accordance with the cue word to unfold, after the collation of the complex prediction output example table is shown in Table 3 (the results are all generated by the Large Language Model; ChatGpt prediction).

4.2. Performance of large language models for profitability forecasting

When evaluating the predictive ability of a large language model, Precision, Recall, and F1 value (F1) are usually chosen for evaluation. Precision measures the accuracy of the model; Recall evaluates the model's ability to recognize positive class samples and represents the coverage ability of the model; F1 value is the reconciled average of the predicted value and recall, which is used to measure the accuracy and completeness of the model in a comprehensive way, and the value of F1 value ranges from 0 to 1. The formulas for accuracy, recall and F1 values are shown in Equation (7), Equation (8), and Equation (9), respectively.

$$\text{Precision} = \frac{TP}{TP + FP} \quad (7)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (8)$$

Table 2
Examples of simple prediction results.

Deadline for projections	Forecasting the direction of earnings
2012	rise
2013	rise
2014	rise
2015	minimize
2016	rise
2017	rise
2018	rise
2019	minimize
2020	rise
2021	rise
2022	rise
2023	rise

$$F1 = 2 * \frac{\text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}} \quad (9)$$

Where TP is the true case, the number of samples correctly predicted by the model as positive classes; FP is the false positive case, the number of samples incorrectly predicted by the model as positive classes; and FN is the false negative case, the number of samples of actual positive classes incorrectly predicted by the model as negative classes. The results of accuracy, recall and F1 values for simple prediction are shown in Fig. 2, the results of accuracy, recall and F1 values for analyst prediction are shown in Fig. 3, and the results of accuracy, recall and F1 values for complex prediction are shown in Fig. 4.

Comparing the two large language model prediction models and the analyst prediction results, the complex prediction has the highest accuracy rate of 64.35 %, indicating that the complex prediction has the best classification performance on all data points. Simple prediction had an accuracy of 60.24 %, which was slightly lower than the complex prediction model, but the difference was not significant and still a stable performance. The Human Analyst Prediction has the lowest accuracy rate of 58.37 %, which is less effective at classifying the overall data and may have more misclassifications. The recall of the human analyst prediction is 84.03 %, which means that the human analyst is able to identify more positive class samples and is suitable for tasks that are very sensitive to the omission of positive class samples. While the recall of simple prediction and complex prediction is relatively lower than the human analyst prediction result, which indicates that the large language model is more conservative in recognizing positive class samples and may have more underclassification. The F1 value of 69.60 % for complex prediction is the highest among the three models, indicating that it has found a better balance between accuracy and recall and has the most superior overall performance. Analyzing the three indicators together, Complex Prediction has higher accuracy (64.35 %), better recall (70.73 %), and the highest F1 value (69.60 %). Taken together, it achieves a better balance between accuracy, recall and F1 value, and is suitable for those tasks that require a balance between the overall classification effect and the ability to recognize positive classes, and is therefore the optimal choice.

4.3. Portfolio assessment based on forecasts from a large language model

Based on the results of the complex prediction of the large language model, considering the three results of predicting the direction of earnings in 2023, the magnitude of the predicted increase or decrease in earnings, and the confidence score of the large language model, 20 companies are randomly selected from the screened results that meet the requirements, and Table 4 lists the 10 companies expected to have earnings growth in 2023, and Table 5 lists the 10 companies expected to have earnings decline in 2023, as the The information set is inputted into the momentum strategy model, after simulated investment, compared with the benchmark market and backtested to obtain the index data in Table 6. The cumulative return curve of the portfolio versus the benchmark index is shown in Fig. 5, the cumulative return curve of rising stocks is shown in Fig. 6, the cumulative return curve of declining stocks is shown in Fig. 7, and the comparison of the daily return of the portfolio versus the benchmark index is shown in Fig. 8, and the distribution of the returns is shown in Fig. 9.

From the results of the portfolio evaluation index, the Sharpe Ratio of the portfolio based on the prediction results of the Large Language Model; is 1.5959. Usually, a Sharpe Ratio of more than 1 is considered to be a better Sharpe Ratio, which means that the portfolio has obtained a superior return under the risk it bears. The maximum retracement is -0.0937, which represents the historical maximum retracement of 9.37 %. Although the retracement is relatively large, it has not exceeded 10 %, so the retracement of 9.37 % is relatively moderate and acceptable. alpha indicates that the portfolio has achieved an excess return of 17.14 % compared to the benchmark. Generally speaking, a higher alpha value

Table 3

Example table of complex prediction results (ChatGpt prediction generation).

mandates	in the end			
Task 1: Identify financial statements	Identify balance sheet, income statement through 2019, 2020			
Task 2: Calculate financial ratios and give the procedure for calculating them	financial ratio	formulas	Calculation process for 2019	Calculation process for 2020
	current ratio	Current ratio = total current assets ÷ total current liabilities	941,698,622.45 ÷ 860,061,158.06 = 1.095	974,154,380.67 ÷ 933,310,018.34 = 1.043
	quick ratio	Quick ratio = (total current assets – inventories) ÷ total current liabilities	(941,698,622.45—379,803,711.01) ÷ 860,061,158.06 = 0.655	(974,154,380.67—388,318,743.34) ÷ 933,310,018.34 = 0.626
	gearing	Gearing ratio = total liabilities ÷ total assets	4,940,966,000 ÷ 8,626,802,000 = 0.57	5,767,134,000 ÷ 9,240,911,000 = 0.62
	net profit margin	Net Profit Margin = Net Profit ÷ Total Operating Revenue x 100 %	558,560,005.13 ÷ 2,149,515,000 = 26.04 %	627,312,724.62 ÷ 2,183,736,000 = 28.74 %
	return on total assets	Return on Total Assets = Net Profit ÷ Total Assets x 100 %	558,560,005.13 ÷ 8,626,802,000 = 6.47 %	627,312,724.62 ÷ 9,240,911,000 = 6.79 per cent
	Return on shareholders' equity	Return on shareholders' equity = Net profit ÷ Shareholders' equity x 100 %	558,560,005.13 ÷ 1,435,019,000 = 38.99 %	627,312,724.62 ÷ 1,566,745,000 = 40.01 %
	gross margin	Gross profit margin = (Revenue from main business – Cost of main business) ÷ Revenue from main business x 100 %	2,138,571,000—1,355,753,000) ÷ 2,138,571,000 = 36.54 %	(2,172,587,000—1,394,766,000) ÷ 2,172,587,000 = 36.00 %
	Inventory turnover	Inventory Turnover = Operating Costs ÷ Inventory	1,357,690,000 ÷ 379,803,711.01 = 3.57	1,396,706,000 ÷ 388,318,743.34 = 3.59
	Accounts receivable turnover ratio	Accounts Receivable Turnover = Operating Income ÷ Accounts Receivable	2,149,515,000 ÷ 671,291,619.99 = 3.20	2,183,736,000 ÷ 671,048,171.00 = 3.25
	earnings per share	Earnings per share = net income ÷ number of common shares	558,560,005.13 ÷ 121,212,121.21 = 4.61	627,312,724.62 ÷ 125,000,000 = 5.02
Task 3: Explain the economic meaning of financial ratios	current ratio	A comparison of the current ratio for 2019 and 2020 shows the change in the company's short-term solvency. The increase in the current ratio indicates that the company has more current assets to meet its short-term liabilities in 2020 compared to 2019, reflecting the company's increased short-term financial security.		
	quick ratio	The change in quick ratio indicates the liquidity of the company after deducting inventories.The improvement in quick ratio in 2020 as compared to 2019 indicates that the quality of the company's current assets has improved, reducing the pressure on funds tied up in inventories.		
	gearing	An increase in gearing indicates that the company is relying on an increased percentage of debt financing in 2020 compared to 2019, which could indicate that the company is taking on a higher level of financial risk, but it could also mean that the company is taking on more leverage in its expansion.		
	net profit margin	The increase in net profit margin indicates the company's profitability as shown by the increase in profit generated per unit of operating income in 2020 as compared to 2019.		
	return on total assets	An increase in the return on total assets implies an increase in the profit that the company earns through its total assets, indicating an improvement in the efficiency of the company in terms of asset utilization.		
	Return on shareholders' equity	The rise in ROE indicates that the company is generating more profits for shareholders in 2020 compared to 2019, reflecting the enhanced efficiency in the utilization of shareholders' equity.		
	gross profit margin	The improvement in gross operating margin suggests that the company's profitability from its main business has increased in 2020 compared to 2019, and in particular, the control of its main costs may have improved.		
	Inventory turnover	The improvement in inventory turnover means that the company has a faster flow of inventory in 2020 compared to 2019, showing an improvement in inventory management.		
	Accounts receivable turnover ratio	The increase in accounts receivable turnover means that the company will be able to collect payments faster in 2020 compared to 2019, improving the efficiency of liquidity utilization.		
	earnings per share	Earnings per share growth reflects the company's improved profitability, with each share generating more profit for shareholders, indicating that the company is operating more effectively in 2020 compared to 2019.		
Task 4: Forecasting the direction of earnings for the coming year	rise			
Task 5: Scoring of predicted information	0.85			
Task 6: Forecasting the magnitude of change in earnings over the next year	Moderate change			
Task 7: Generate an analytical report	1. Financial ratio analysisCurrent ratio is an important measure of a company's short-term solvency, reflecting the company's ability to repay its current liabilities from current assets.The company's current ratio in 2019 is 1.09, indicating that the company is in a relatively safe position in terms of short-term debt servicing. However, the current ratio declined slightly to 1.04 in 2020, and while it is still above 1, implying that the company is still able to cope with its short-term liabilities relatively well, there is a decrease in liquidity, which may reflect that the company has increased some of its short-term liabilities in 2020 or that the growth rate of its current assets is lower than the growth rate of its liabilities. The slight decline in the liquidity ratio, while increasing the pressure on short-term debt servicing to some extent, still maintains a relatively robust level overall. The quick ratio is a liquidity metric, net of inventory, that more accurately reflects a company's cash liquidity for short-term debt repayment.The company's quick ratio is 0.65 in 2019, and declines slightly to 0.63 in 2020. Although the decrease in the quick ratio implies a weakening of the company's short-term debt service capacity, the difference between the two years is not significant. Considering that the quick ratio is below 1, it indicates that the company may face some liquidity pressure in the short term, especially in the case of emergencies where inventory is difficult to liquidate quickly. Therefore, the company may need to further optimize its accounts receivable and inventory management in the future to ensure the quality and liquidity of its current assets.Gearing ratio reflects the proportion of a company's total assets that are financed by debt, revealing the company's financial risk.In 2019 the company's gearing ratio is			

(continued on next page)

Table 3 (continued)

mandates	in the end
	0.57, while in 2020 the ratio rises to 0.62, an increase of 5 percentage points. This change indicates that the company is more dependent on debt financing in 2020 compared to 2019. This could be a result of an increase in the source of funds required for the company to expand or make large-scale investments, which could lead to an increase in the gearing ratio. While debt financing can lead to more capital and resources, it can also increase a company's financial risk. A higher level of debt may mean that the company has to incur greater interest expenses in servicing its debt, especially in times of economic volatility or changes in market conditions, which may affect the company's cash flow and profitability. Net Profit Margin is a key measure of a company's profitability, which indicates the net profit that the company is able to earn per unit of operating revenue. In 2019 the company's Net Profit Margin is 26.04 %, while in 2020 this indicator rises to 28.74 %. The increase in net profit margin indicates that the company has made significant progress in controlling costs and increasing sales revenue. In 2020, the company was able to convert revenue growth into profit more efficiently, showing improvements in operational efficiency and cost control. The increase in net profit margin not only strengthens the Company's profitability, but also provides more adequate financial support for future expansion and investment. Return on Total Assets (ROA) is a measure of how efficiently a company utilizes its assets, reflecting how the company generates profits from its total assets. The company's ROA is 6.47 % in 2019, while it rises to 6.79 % in 2020. The increase in ROA suggests that the company has improved the efficiency of its asset utilization, and has been able to generate more profits by investing in fewer assets. Although the increase in ROA is small, this change indicates that the company has optimized its resource allocation and asset utilization, enhancing its overall profitability. The company has effectively increased the value-added effect of its assets by improving production efficiency and optimizing investment decisions. Return on Shareholders' Equity (ROE) is a measure of a company's ability to generate returns for its shareholders. The Company's ROE for 2019 is 38.99 %, while it improves to 40.01 % in 2020. The improvement in ROE reflects the Company's better performance in utilizing shareholders' investment, and the return on shareholders' investment further increases. In particular, against the backdrop of fierce market competition and uncertain economic environment, the ability to increase the return on shareholders' equity indicates that the Company has performed better in terms of operational efficiency, capital utilization and profit distribution, which has enhanced shareholders' confidence. Gross operating margin is a key measure of the profitability of a company's primary business. The company's gross operating margin was 36.54 % in 2019 compared to 36.00 % in 2020. Although the gross profit margin has decreased, the decrease is small and remains at a relatively healthy level. The stability of the gross operating margin indicates that the company is performing better in controlling the cost of its main business and boosting its sales revenue. Small fluctuations in gross profit margin may be related to factors such as changes in raw material costs and price adjustment strategies, but overall, the Company's profitability in its main business remains strong. Inventory turnover ratio reflects the liquidity and saleability of the company's inventory. the inventory turnover ratio is 3.57 in 2019 and rises slightly to 3.59 in 2020. this change indicates that the company has improved its inventory management, with faster inventory turnover and shorter time of capital utilization. The improvement in inventory turnover ratio means that the company can manage its inventory more effectively in the course of its operations, reduce inventory backlog and capital utilization, and help to improve the liquidity of the enterprise and the efficiency of capital utilization. The accounts receivable turnover ratio reflects how efficiently a company collects its receivables. the company's accounts receivable turnover ratio is 3.20 in 2019 and rises slightly to 3.25 in 2020. the increase in accounts receivable turnover ratio implies that the company collects more efficiently in 2020, which reduces the risk of bad debts and capital tie-ups, and further improves the company's cash flow and turnover. the company's accounts receivable turnover ratio is 3.20 in 2019 and increases slightly to 3.25 in 2020. Earnings per share is a measure of the net profit a company can earn per share of stock. Earnings per share are \$4.61 in 2019 and \$5.02 in 2020, an increase of 8.9 %. Earnings per share growth is a direct reflection of the company's increased profitability and the creation of more value for shareholders. The company's earnings per share growth demonstrates its success in revenue growth, cost control and profit growth, and enhances investor confidence. 2. Projections of future earnings Based on the company's financial performance in recent years, we forecast that the company will continue its favorable profitability trend in the coming year. First, the improvement in profitability metrics such as net profit margin, return on total assets, and return on shareholders' equity indicates the company's continued improvement in profitability and resource utilization. Second, the increase in inventory turnover and accounts receivable turnover indicates that the company has optimized its capital management and asset liquidity and is able to use capital more efficiently. As a result, we expect earnings to show growth in the coming year. However, the increase in gearing ratio also hints at potential financial risks, especially in the event of rising interest rates or a volatile economic environment, where a higher debt burden may put some pressure on the company. As a result, despite the company's growth potential, the magnitude of change in earnings is expected to be moderate and may face some volatility. 3. Conclusion Overall, the company's change in financial ratios between 2019 and 2020 shows solid growth momentum. Improved profitability, more efficient use of assets, and increased returns to shareholders all provide strong support for future earnings growth. However, higher debt ratios and slightly declining liquidity metrics still remind the company of the need to maintain a prudent financial management strategy. Taking these factors into account, the Company's earnings are expected to continue to grow in the coming year, but may experience some volatility, especially in light of changes in the external economic environment and internal financial structure. In order to maintain sustained growth, the Company should continue to optimize its debt structure and strengthen its capital management to ensure financial soundness and maximize shareholder returns.

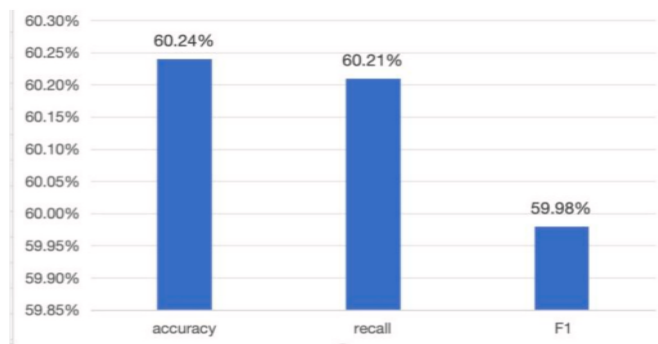


Fig. 2. Simple prediction accuracy, recall value, F1 value.

means that the portfolio was able to achieve a return that exceeded the market average through stock selection or market timing. 17.14 % is already a high return compared to the benchmark. The 17.14 % is already a good value of excess return, which indicates that the return of the portfolio predicted by the Large Language Model; is relatively good. β value is 0.4769, which indicates that the volatility of the portfolio is only 47.69 % of the market, which is relatively low-risk, and therefore

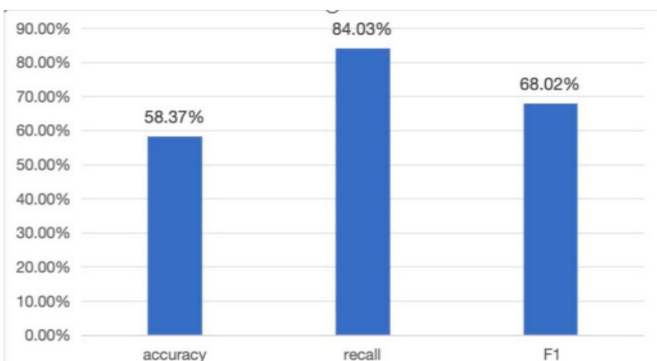


Fig. 3. Analysts measure accuracy, recall value, F1 value.

the portfolio is more suitable for conservative investors. The information ratio of 0.5311, which measures the excess return per unit of tracking error, is within reasonable range. Combining the results of all evaluation metrics, this portfolio is a moderate risk with good returns, with the only shortcoming being the historical maximum retracement magnitude.

From the benefit curve of the portfolio, the cumulative return curve

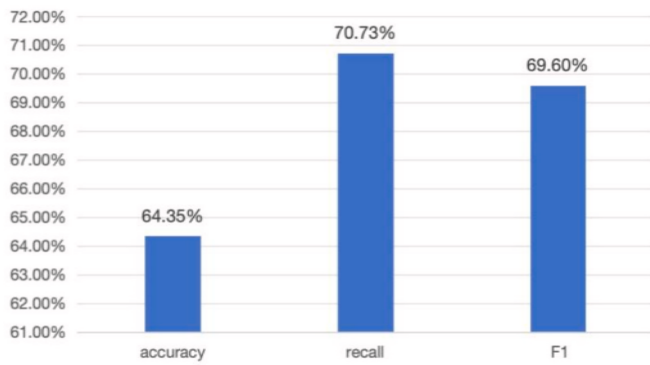


Fig. 4. Complex prediction accuracy, recall value, F1 value.

Table 4

Table of Companies Expected to Grow Profitably in 2023.

Company Code	Projected earnings direction for 2023	Estimated change in earnings	Predicting informativeness
000,568	rise	huge changes	0.85
002,146	rise	huge changes	0.85
002,157	rise	huge changes	0.85
002,236	rise	huge changes	0.85
002,371	rise	huge changes	0.85
002,380	rise	huge changes	0.85
600,009	rise	huge changes	0.85
600,153	rise	huge changes	0.85
600,519	rise	huge changes	0.85
600,809	rise	huge changes	0.85

Table 5

Table of Companies Expected to Reduce Earnings in 2023.

Company Code	Projected earnings direction for 2023	Estimated change in earnings	Predicting informativeness
002,009	minimize	huge changes	0.85
002,180	minimize	huge changes	0.85
002,258	minimize	huge changes	0.85
002,497	minimize	huge changes	0.85
600,089	minimize	huge changes	0.85
600,389	minimize	huge changes	0.85
600,438	minimize	huge changes	0.85
600,596	minimize	huge changes	0.85
600,990	minimize	huge changes	0.85
601,699	minimize	huge changes	0.85

Table 6

Data table for assessment indicators.

norm	Sharpe ratio	Maximum retracement value	alpha value	beta value	Information ratio
numerical value	1.5959	-0.0937	0.1714	0.4769	0.5311

of the portfolio versus the benchmark index shows the change in the cumulative return of the portfolio and the change in the cumulative return of the benchmark index (CSI 300), and Fig. 5 shows that the cumulative return of the portfolio is gradually higher than that of the benchmark of the CSI 300 index, and the cumulative return of the portfolio has been higher than the benchmark's return after March-April 2023, which demonstrates good excess returns. Fig. 6 Cumulative Return Curve of Rising Stocks and Fig. 7 Cumulative Return Curve of Declining Stocks show in detail the change of each stock's return, which can help managers to reevaluate whether or not to continue to hold the stock at any time, e.g., 601,699 stock among the declining stocks has a



Fig. 5. Cumulative return profile of the portfolio versus the benchmark index.

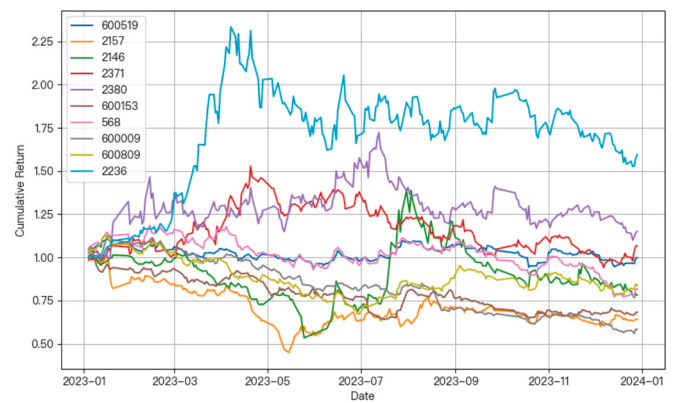


Fig. 6. Cumulative return curve for rising stocks.

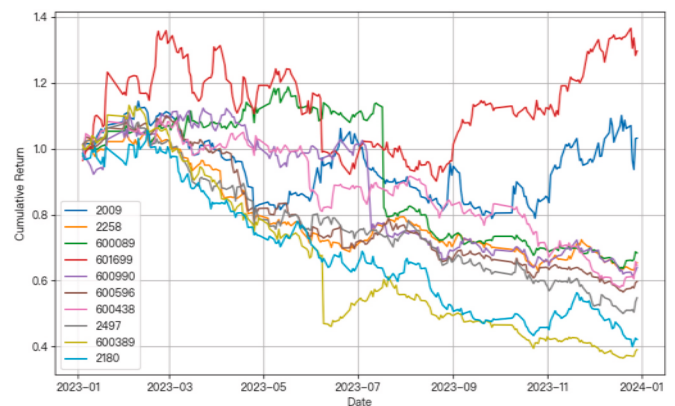


Fig. 7. Cumulative Return Curve for Declining Stocks.

rising return from November-December 2023, and investment managers can change the number of holdings ahead of time to further optimize the investment portfolio. Fig. 8 The daily return curve of the portfolio versus the benchmark index shows the daily change in returns of the portfolio and the market benchmark. Time periods with large gaps between the two curves, such as February, May, July, and November, indicate that the portfolio's daily returns differ more from the benchmark's, suggesting that the portfolio performs better during this time period. Fig. 9 Return Distribution Graph includes two distributions, one for the daily returns of the portfolio and the other for the benchmark index, showing the distribution of the portfolio's daily returns versus the benchmark, which can be used to determine if there is excessive volatility. The

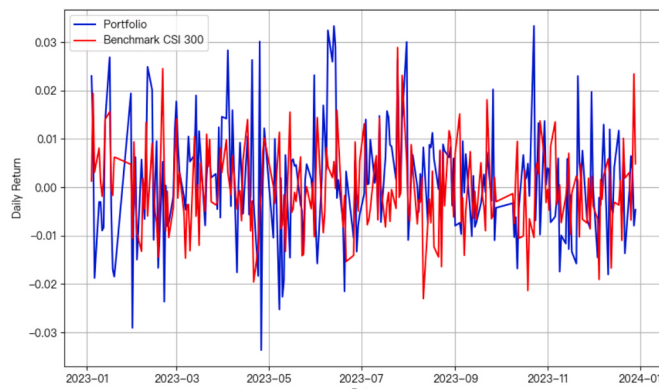


Fig. 8. Portfolio versus benchmark index daily returns.

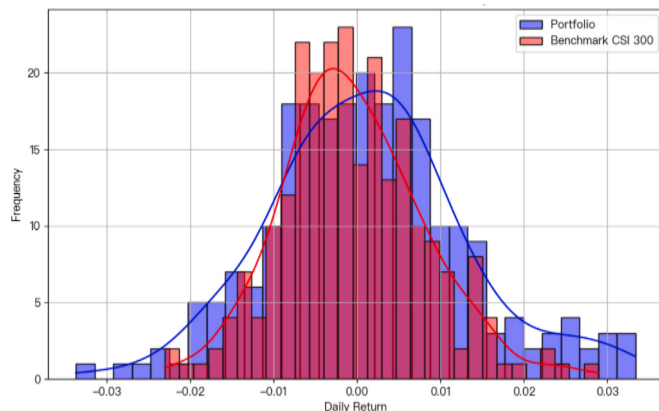


Fig. 9. Distribution of proceeds.

portfolio return distribution is more concentrated in the positive return range than the benchmark return distribution, and the portfolio is more robust and able to consistently generate positive returns.

5. Further discussion and outlook

This paper investigates whether large language models can perform the work of human analysts by constructing simple forecasts based on large language models and complex forecasts prompted by chain of thought to predict the direction of a company's earnings in the coming year. Financial statement analysis is a traditionally quantitative task that requires critical thinking, reasoning, and judgment. The Large Language Model; is provided with standardized and anonymized financial statement data from 608 publicly traded companies and uses chained thinking prompts to simulate the analysis process of a human analyst. The results show that the chain-thinking-prompted analysis of the Large Language Model; provides valuable insightful information that allows the model to outperform professional human analysts in predicting the direction of future earnings. In addition, constructing a momentum strategy for the portfolio based on the complex prediction results of chained thinking and backtesting it based on the historical data of the CSI 300 index, the results show that compared to the benchmark index, the portfolio return rises steadily and is suitable for prudent investors, and is able to consistently generate positive returns. It proves the decision-making ability of the AI Large Language Model; to analyze financial statements in the field of accounting and finance, predict future returns in the field of economics, and perform portfolio optimization. In this study, the fact that the Large Language Model; successfully accomplished a task that normally requires the expertise and judgment of a human analyst, and was able to do so with numerical data alone, is

consistent with the fact that the Large Language Model; has near-human learning and analytical capabilities, and suggests that the Large Language Model; has the potential to liberalize and de-specialize financial information processing. Liberalization refers to the use of large language models can be readily analyzed the company's financial situation and the company's future earnings forecasts, investors can freely access the operating conditions of listed companies. De-specialization refers to the fact that earnings forecasts are no longer dependent on professional financial analysts, making this task de-specialized.

Our findings underscore that the true value of large language models (LLMs) in finance lies not in displacing human analysts or conventional models, but in transforming the nature of financial reasoning through chain-of-thought (CoT)-enhanced automation. By rendering numerical financial logic both transparent and scalable, CoT-augmented LLMs create new opportunities for AI applications across the financial sector, such as real-time financial statement analysis and risk assessment, scenarios in which both high accuracy and interpretability are essential. Future work will focus on refining CoT prompting strategies to accommodate industry-specific financial logics, for example, those distinguishing manufacturing from technology sectors, in order to fully realize this potential. In addition, future research also could be combined with financial statement analysis to study more aspects, such as the output of short analytical reports in the forecast results, which may reflect a sentiment bias or potential bias towards the company's financial position.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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MeiqunYin is the Vice Dean, Professor, Ph.D. supervisor, and Certified Public Accountant (CPA) at the School of Business, China University of Political Science and Law. She obtained her Ph.D. in Accounting from Renmin University of China in 2005. Professor Yin has extensive experience in both academic and administrative management roles and has held several positions, including serving as a regional manager for the Beijing 2008 Olympic Games. She has contributed to various committees, including the Chinese Accounting Association and the IMA China Management Accounting Expert Committee. She has received numerous awards for her academic research and teaching, including the First Prize for Research Achievement from Beijing Second Foreign Language University and the “Young and Middle-aged Backbone Teacher” title. Additionally, she has received several national research funds. Professor Yin has made significant contributions to accounting education and public service. Email: yinmeiqun@126.com

Mengzhu Guo(corresponding author), a dedicated professional with a strong academic and practical background. She earned her Bachelor's degree in Accounting from Northeastern University (2014–2018), followed by a Master's degree in Information Management from The Chinese University of Hong Kong (2018–2019). Currently, she is pursuing a Doctoral degree in Theoretical Economics at China University of Political Science and Law. Dr. Guo has published extensively in Chinese journals, focusing on economic forecasting, new-quality productivity, and the development of evaluation indicators for economic progress in China. She has significant experience in the financial sector, having worked in supply chain finance and debt resolution at China Railway Commercial Factoring Co., Ltd. Her expertise also includes financial optimization and project management. Email: gmengzhu@126.com